PMA 159 Verbal Test

Complete Study Guide - Formulas, Methods, Shortcut Tricks & Solved Examples

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1 Age Problem

KEY FORMULAS / FACTS

- Age after n years = Present age + n
- Age n years ago = Present age - n
- The **difference** between two ages **never changes**
- If ratio A:B = a:b then A = a x, B = b x (x = 1 unit)
- Sum of ages after n years = Present sum + (n x number of people)
- Average age = Sum of ages / Number of people

STANDARD METHOD

1. Assume present ages as **x** (or as ratio units **ax, bx**).
2. Write the condition for past/future as an equation: (age + n).
3. Solve for x, then answer exactly what is asked (present / past / future?).
4. Verify by putting the numbers back into the story.

SHORTCUT TRICK

Ratio-Unit Trick: Convert the ratio into units, find the value of 1 unit from the given sum or difference, and multiply. **Difference-Lock:** the gap between ages is constant, so use the gap to jump from past to future.

Memory Hook: D.U.S. = **D**ifference stays, **U**nit method, **S**um grows by n x people.

When	Person A	Person B
Now	x	y
After n yrs	x + n	y + n
n yrs ago	x - n	y - n
Difference	y - x	y - x (same!)

SOLVED EXAMPLES

Q1. Father is 3 times his son's age. After 10 years the father will be twice the son's age. Find present ages.

Solution: Son = S, Father = 3S. 3S + 10 = 2(S + 10) → 3S + 10 = 2S + 20 → S = 10.

Answer: Son 10 yrs, Father 30 yrs

Q2. The ages of A and B are in ratio 5:7 and their sum is 48. Find A's age.

Solution: 5x + 7x = 48 → 12x = 48 → x = 4. A = 5 × 4 = 20.

Answer: A = 20 (B = 28)

Q3. 5 years ago A was twice as old as B. Now the sum of their ages is 40. Find A's present age.

Solution: A - 5 = 2(B - 5) → A = 2B - 5. Put in A + B = 40: 3B - 5 = 40 → B = 15, A = 25.

Answer: A = 25 years

2 Direction Sense

N

KEY FORMULAS / FACTS

- Clockwise order: **N → E → S → W → N**
- Right turn = 90 deg clockwise; Left turn = 90 deg anti-clockwise
- Distance = $\sqrt{(a^2 + b^2)}$ for right-angle paths (3-4-5, 5-12-13, 8-15-17)
- Net displacement: add opposite directions with sign (N = +, S = -, E = +, W = -)
- **Morning:** Sun in East, shadow falls **West**. **Evening:** Sun in West, shadow falls **East**
- Facing the Sun at sunrise = facing East; at sunset = facing West

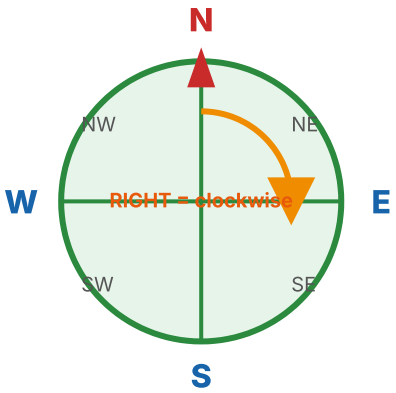
STANDARD METHOD

1. Start at a point and draw an arrow for every move on a small sketch.
2. Mark turns carefully: look at the way you are **currently facing**.
3. Cancel opposite movements (N vs S, E vs W) to get net X and Y.
4. Use Pythagoras for the shortest distance; use the final X/Y signs for direction.

SHORTCUT TRICK

Hand-Compass Trick: Hold your hand flat - after a right turn you rotate one step clockwise on NESW; after a left turn one step anti-clockwise. For shadows: **Morning shadow = West; Evening shadow = East.**

Memory Hook: "Never Eat Soggy Waffles" = N, E, S, W clockwise. Right = move forward on the list, Left = move back.



SOLVED EXAMPLES

Q1. A man walks 10 km North, turns right and walks 6 km, then turns right again and walks 10 km. How far and in which direction is he from the start?

Solution: North 10, East 6, then facing East turns right → South 10. N and S cancel. Net = 6 km East.

Answer: 6 km East

Q2. Rohan walks 3 km West, then turns left and walks 4 km. Find the distance from the start point.

Solution: Facing West, left turn = South. Displacement: 3 W and 4 S. Distance = $\sqrt{(3^2+4^2)} = 5$.

Answer: 5 km (South-West)

Q3. One morning, Amit's shadow falls to his right. Which direction is he facing?

Solution: Morning shadow falls West, so West is on his right. Right of South is West. So he faces South.

Answer: South

3 Percentage

%

KEY FORMULAS / FACTS

- x% of y = (x x y) / 100
- Percentage = (Part / Whole) x 100
- % Change = (Change / Original) x 100
- Successive change: Net % = a + b + (ab/100) (use - for a decrease)
- If A is x% more than B, then B is **x/(100+x) x 100**% less than A
- Population/Value after n years = $P\ (1 + r/100)^n$

STANDARD METHOD

1. Convert the percent to a fraction or to /100.
2. Identify the BASE (the number after 'of' / 'than' / 'is').
3. Multiply or divide by the base; for change problems use (new - old)/old.
4. For 'x% more/less' compare to the original, never to the new value.

SHORTCUT TRICK

Fraction Table Trick: Memorise the fraction for each common percent; 25% = 1/4 so just divide by 4! **Swap Trick:** a% of b = b% of a (8% of 50 = 50% of 8 = 4).

Memory Hook: "1-2-3-4-5-6-7-8-9-10" as the denominators: 100, 50, 33.3, 25, 20, 16.6, 14.3, 12.5, 11.1, 10 %.

Fraction	%	Fraction	%
1/2	50%	1/6	16.67%
1/3	33.33%	1/7	14.28%
1/4	25%	1/8	12.5%
1/5	20%	1/9	11.11%
2/3	66.67%	1/10	10%
3/4	75%	1/12	8.33%

SOLVED EXAMPLES

Q1. Find 20% of 450.

Solution: 20% = 1/5, so 450 / 5 = 90.

Answer: 90

Q2. The price of a shirt is increased by 20% and then decreased by 20%. Find the net change.

Solution: Net = 20 - 20 + (20 x -20)/100 = -4%.

Answer: 4% decrease

Q3. A's salary is 25% more than B's. By what percent is B's salary less than A's?

Solution: 25/(100+25) x 100 = 20%.

Answer: 20% less

Q4. A student scores 40% and fails by 20 marks. Another scores 50% and passes by 40 marks. Find the total marks.

Solution: Difference of 10% = 20 + 40 = 60 marks, so 100% = 600.

Answer: 600 marks

4 Ranking & Comparison

KEY FORMULAS / FACTS

- **Total = Rank from Top + Rank from Bottom - 1**
- Rank from other end = Total + 1 - Given rank
- People between A and B (same side) = |Rank A - Rank B| - 1
- People between (opposite sides) = Total - (Rank A from top + Rank B from bottom)
- Tallest/heaviest/oldest vs shortest/lightest/youngest: arrange in order, then compare
- 'A is taller than B' means A > B; write with > signs

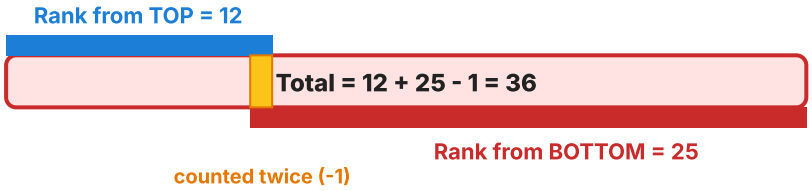
STANDARD METHOD

1. Draw a line of positions. Mark known positions.
2. For comparisons write the order with > signs: A > B > C.
3. Merge statements by joining the common person.
4. Apply the Total formula or the 'between' formula.

SHORTCUT TRICK

The Overlap Trick: Top + Bottom counts the person once too many, so always **subtract 1**. For two people swapping, the first person's NEW position = the second person's OLD position.

Memory Hook: "T + B - 1" = **T**otal, **B**ottom... you always lose 1 (the person counted twice).



SOLVED EXAMPLES

Q1. Rahul is 12th from the top and 25th from the bottom in a class. How many students are there?

Solution: Total = 12 + 25 - 1 = 36.

Answer: 36

Q2. In a row of 40 students, Ali is 15th from the left. What is his position from the right?

Solution: 40 + 1 - 15 = 26.

Answer: 26th from right

Q3. A is 7th and B is 18th from the top. How many students are between them?

Solution: 18 - 7 - 1 = 10.

Answer: 10

Q4. Ram is 10th from the left and Shyam is 15th from the right. After exchanging places, Ram is 20th from the left. Find the total.

Solution: Ram now sits at Shyam's old place = 20th from left. Total = 20 + 15 - 1 = 34.

Answer: 34

5 Coding & Decoding

KEY FORMULAS / FACTS

- Letter value: A=1, B=2 ... Z=26
- **Opposite letter = 27 - position** (A↔Z, B↔Y ...)
- Types: **Letter shift (+/-n)**, **Reverse order**, **Substitution**, **Number/Symbol code**, **Sum of positions**
- Reverse-alphabet (opposite) code: CAT → XZG
- Word-to-word code: compare common words in two sentences

STANDARD METHOD

1. Write positions of every letter under the word.
2. Compare the coded word letter by letter: find the +/- shift or the opposite.
3. Check if the shift is the same for every letter or increases (+1, +2, +3).
4. Apply the same rule to the new word.

SHORTCUT TRICK

EJOTY + Mirror Trick: Remember E=5, J=10, O=15, T=20, Y=25 as anchors. To find the opposite letter, subtract position from 27, or just use the A-M / Z-N table below.

Memory Hook: "EJOTY" (say it like 'Ee-Joe-Tie') = positions 5, 10, 15, 20, 25.

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
Z	Y	X	W	V	U	T	S	R	Q	P	O	N
26	25	24	23	22	21	20	19	18	17	16	15	14

Each vertical pair adds up to 27 (A+Z, B+Y ... M+N). Highlighted = **E J O T Y** (5, 10, 15, 20, 25).

SOLVED EXAMPLES

Q1. If MANGO is coded as NBOHP, how is PEACH coded?

Solution: Every letter +1: P→Q, E→F, A→B, C→D, H→I.

Answer: QFBDI

Q2. Find the opposite (mirror) letter of G.

Solution: 27 - 7 = 20 → T.

Answer: T

Q3. In a code, 'Red' means 'Blue', 'Blue' means 'Green', 'Green' means 'Yellow'. What is the colour of the sky?

Solution: Sky is actually Blue, and Blue is coded as Green.

Answer: Green

Q4. If A=1, B=2, C=3 ... what is the value of CAB?

Solution: C=3, A=1, B=2 → sum = 6 (or code 312).

Answer: 6 / 312 (as the question rule)

6 Water & Mirror Image

M

KEY FORMULAS / FACTS

- **Mirror image** = left-right flip (like a mirror to the side). Letters/words appear **reversed in order**.
- **Water image** = top-bottom flip (reflection in water below). Letters stay in the same order.
- **Clock - Mirror time = 11:60 - T** (Example: 3:25 → 8:35)
- **Clock - Water time = 6:00 - T** (Example: 4:30 → 1:30)
- If time is exactly on the hour: Mirror = 12 - T (3:00 → 9:00)
- Letters symmetrical about a vertical axis (look same in mirror): **A H I M O T U V W X Y**
- Letters symmetrical about a horizontal axis (same in water): **B C D E H I K O X**

STANDARD METHOD

1. Decide: Mirror (left-right) or Water (top-bottom)?
2. **Words:** Mirror → reverse the order of letters and flip each. Water → same order, flip each upside-down.
3. **Clock:** use the subtraction formula, then adjust (if you get a negative or 12+).
4. Always cross-check with a small sketch of the hands.

SHORTCUT TRICK

Subtraction Trick: Mirror = **11:60 - time**; Water = **6:00 - time**. Also **H, I, O, X** look identical in both mirror and water - they are the safe letters.

Memory Hook: "HIOX" = H-I-O-X never change. **"M for Mirror = 11:60"**, **"W for Water = 6:00"** (6 looks like a W turned round).



SOLVED EXAMPLES

Q1. Mirror image of the time 3:25.

Solution: 11:60 - 3:25 = 8:35.

Answer: 8:35

Q2. Water image of the time 4:30.

Solution: 6:00 - 4:30 = 1:30.

Answer: 1:30

Q3. Mirror image of the word 'MAT' (all letters symmetrical).

Solution: Reverse the order: T, A, M. All letters are symmetric so they look the same.

Answer: TAM

Q4. Water image of 'BOX'.

Solution: B, O and X are all symmetrical about a horizontal axis; order stays the same.

Answer: BOX (unchanged)

7 Maths Series

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KEY FORMULAS / FACTS

- **AP:** nth term = a + (n - 1)d; Sum = n/2 [2a + (n - 1)d]
- **GP:** nth term = a r^{n-1}; ratio r = next / previous
- Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 ... Cubes: 1, 8, 27, 64, 125, 216 ...
- Fibonacci: each term = sum of previous two (1,1,2,3,5,8,13)
- Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
- **Second-order difference:** if differences themselves form an AP, take differences of differences
- Alternating series: two interleaved series (odd and even positions)

STANDARD METHOD

1. Find the **difference** between terms. Same? → AP.
2. If not, find the **ratio** → GP.
3. If neither, check differences of differences, squares/cubes ($n^2 \pm k$), primes, Fibonacci.
4. If still stuck, split into alternate terms (odd/even positions).

SHORTCUT TRICK

D-R-S-A Check: Difference → **R**atio → **S**quare/cube → **A**lternate. Also memorise squares up to 20 (11^2=121, 12^2=144, 13^2=169, 14^2=196, 15^2=225, 16^2=256, 17^2=289, 18^2=324, 19^2=361, 20^2=400).

Memory Hook: "DRSA" - 'Don't Rush, Study Again'. Always try them in that order.

Pattern	Series	Next
+d every time	2, 5, 8, 11, ?	14 (AP)
x r every time	3, 6, 12, 24, ?	48 (GP)
n^2	1, 4, 9, 16, 25, ?	36
Prime	2, 3, 5, 7, 11, ?	13
Sum of last two	1, 1, 2, 3, 5, 8, ?	13
$n^2 + 1$	2, 5, 10, 17, 26, ?	37

SOLVED EXAMPLES

Q1. 2, 6, 12, 20, 30, ?

Solution: Differences: 4, 6, 8, 10 → next difference 12. 30 + 12 = 42.

Answer: 42

Q2. 5, 10, 20, 40, 80, ?

Solution: Ratio is 2 each time. 80 × 2 = 160.

Answer: 160

Q3. 1, 3, 6, 10, 15, ?

Solution: Differences: 2, 3, 4, 5 → next 6. 15 + 6 = 21 (triangular numbers).

Answer: 21

Q4. 3, 8, 15, 24, 35, ?

Solution: These are $n^2 - 1$ for n = 2,3,4,5,6 → $7^2 - 1 = 48$.

Answer: 48

8 Analogy

KEY FORMULAS / FACTS

- **A : B :: C : D** means (relation of A to B) = (relation of C to D)
- Common relations: Tool - Work, Worker - Workplace, Animal - Young one, Part - Whole
- Cause - Effect, Synonym, Antonym, Degree (small - large), Gender (male - female)
- Number analogy: square / cube / multiply / add pattern
- Letter analogy: shift of positions, reverse, opposite

STANDARD METHOD

1. Read the first pair and find the exact relation.
2. Form a **sentence**: 'A is used for B' or 'B is the young one of A'.
3. Apply the same sentence to the second pair.
4. Eliminate options that do not satisfy the same sentence.

SHORTCUT TRICK

Sentence Trick: Convert the first pair into one clear sentence (e.g. 'A *is a tool to* B'). Then test every option with **the same sentence**. If the sentence fails, the option is wrong.

Memory Hook: "A is to B as C is to ?" - say the sentence out loud, then fill the blank.

Relation	Example 1	Example 2
Tool : Work	Pen : Write	Knife : Cut
Young one	Cow : Calf	Dog : Puppy
Place : Worker	Hospital : Doctor	School : Teacher
Part : Whole	Petal : Flower	Page : Book
Cause : Effect	Fire : Smoke	Rain : Flood
Number	5 : 25	6 : 36 (square)

SOLVED EXAMPLES

Q1. Doctor : Hospital :: Teacher : ?

Solution: Doctor works in a Hospital → Teacher works in a School.

Answer: School

Q2. Pen : Write :: Knife : ?

Solution: A pen is used to write → a knife is used to cut.

Answer: Cut

Q3. Cow : Calf :: Dog : ?

Solution: Calf is the young one of the cow → Puppy for a dog.

Answer: Puppy

Q4. 5 : 25 :: 7 : ?

Solution: Square of number: $5^2 = 25 \rightarrow 7^2 = 49$.

Answer: 49

9 Classification (Odd One Out) X

KEY FORMULAS / FACTS

- Find the **common property** of the group; the one that breaks it is the answer.
- Types: **Word** (category), **Number** (prime, square, even, multiple), **Letter** (positions/gap), **Mixed**
- Number tests: prime? perfect square? multiple of k? sum of digits?
- Letter tests: gap between letters, vowels/consonants, reverse pairs

STANDARD METHOD

1. Read all options; look for 3 that share a property.
2. Test quickly: category, number property, letter gap, size/shape.
3. The leftover item is the odd one out.
4. If two ideas work, choose the more specific/obvious property.

SHORTCUT TRICK

Group-of-3 Trick: In 4 options always search for the property shared by **three**. Check in this order: **Category → Number property → Spelling/letters → Use/function.**

Memory Hook: "3 are twins, 1 is the stranger."

Type	Options	Odd one
Words	Apple, Mango, Potato, Banana	Potato (vegetable, others fruits)
Numbers	2, 3, 5, 9, 11	9 (not prime)
Squares	16, 25, 36, 48	48 (not perfect square)
Animals	Tiger, Lion, Cow, Leopard	Cow (not carnivore)
Letters	ACE, GIK, MOQ, PRS	PRS (gap not 1 letter)

SOLVED EXAMPLES

Q1. Apple, Mango, Potato, Banana

Solution: Apple, Mango and Banana are fruits; Potato is a vegetable.

Answer: Potato

Q2. 2, 3, 5, 9, 11

Solution: 2, 3, 5, 11 are prime; $9 = 3 \times 3$.

Answer: 9

Q3. 16, 25, 36, 48

Solution: $16 = 4^2$, $25 = 5^2$, $36 = 6^2$, but 48 is not a square.

Answer: 48

Q4. ACE, GIK, MOQ, PRS

Solution: In ACE, GIK, MOQ each letter skips one letter (+2). PRS goes +2 then +1.

Answer: PRS

10 Expression / Idioms & Phrases

Aa

KEY FORMULAS / FACTS

- An **idiom** has a meaning different from its words (literal meaning is wrong).
- A **phrasal verb** = verb + preposition (give up, call off, look after).
- Meaning comes from **context**: read the whole sentence first.
- Learn idioms in groups: **Time**, **Money**, **Secrets**, **Trouble**, **Ease/Difficulty**.

STANDARD METHOD

1. Read the sentence and guess the feeling: good, bad, easy, secret, trouble?
2. Eliminate options that match the **literal** meaning.
3. Match the remaining options with the idiom's real meaning.
4. Plug your chosen meaning back in the sentence to check.

SHORTCUT TRICK

Picture + Exaggerate Trick: Imagine the literal picture and make it funny - 'cost an arm and a leg' = you pay with body parts = **very expensive**; 'raining cats and dogs' = animals falling = **heavy rain**. Group by theme to learn 5 idioms at once.

Memory Hook: "Never take it literally" - the picture is the memory, the meaning is the answer.

Idiom	Meaning
Break the ice	Start a conversation in an awkward situation
Once in a blue moon	Very rarely
Piece of cake	Very easy
Spill the beans	Reveal a secret
Hit the nail on the head	Say exactly the right thing
Bite the bullet	Face a hard situation bravely
Burn the midnight oil	Work/study late at night
A blessing in disguise	Bad thing that turns out good
Beat around the bush	Avoid the main point
Cost an arm and a leg	Very expensive
Under the weather	Feeling ill
Let the cat out of the bag	Reveal a secret by mistake
Call it a day	Stop working

Idiom	Meaning
At the eleventh hour	At the last moment
Bolt from the blue	Sudden surprise
Turn a blind eye	Pretend not to see
In hot water	In trouble
Add fuel to the fire	Make a bad situation worse
Apple of one's eye	Very dear person
Cut corners	Do something cheaply/poorly
Look into	Investigate
Put off	Postpone
Call off	Cancel
Give up	Quit
Turn down	Reject
Carry out	Perform / execute

SOLVED EXAMPLES

Q1. 'Break the ice' means:

Solution: Not to smash ice - it means start a friendly conversation.

Answer: Start a conversation

Q2. 'He spilled the beans about the surprise party.'

Solution: Spilled the beans = revealed the secret.

Answer: Revealed the secret

Q3. 'The exam was a piece of cake.'

Solution: Piece of cake = very easy.

Answer: Very easy

Q4. 'The meeting was called off.'

Solution: Called off = cancelled.

Answer: Cancelled

11 Time Sequence & Calendar

T

KEY FORMULAS / FACTS

- **Odd days = (Total days) mod 7** (remainder when divided by 7)
- Ordinary year = 1 odd day; Leap year = 2 odd days
- 100 yrs = 5 odd days; 200 = 3; 300 = 1; 400 = 0 odd days
- Month odd days: 31-day month = 3, 30-day month = 2, Feb = 0 (29 → 1)
- Leap year if divisible by 4 (centuries must be divisible by 400)
- Day number: Sun = 0, Mon = 1, Tue = 2, Wed = 3, Thu = 4, Fri = 5, Sat = 6
- Logical order: **Birth → Childhood → Youth → Adult → Old age** etc.

STANDARD METHOD

1. Convert the gap in days into **odd days** (divide by 7, keep the remainder).
2. Add the odd days to the given day number.
3. If sum > 6, subtract 7. That is the day.
4. For sequences of events: order by the logical or time-based flow.

SHORTCUT TRICK

Mod-7 Wheel: Think of a wheel of 7 days. After 7, 14, 21 ... days you come to the same day. Only the **remainder** matters. For days before, subtract the odd days instead.

Memory Hook: "7 = nothing" - drop every full week, count only what is left.

Sun	Mon	Tue	Wed	Thu	Fri	Sat
0	1	2	3	4	5	6

SOLVED EXAMPLES

Q1. Today is Monday. What day will it be after 100 days?

Solution: 100 mod 7 = 2. Monday + 2 = Wednesday.

Answer: Wednesday

Q2. The day before yesterday was Thursday. What day will it be the day after tomorrow?

Solution: Day before yesterday = Thu, so today = Sat. Day after tomorrow = Sat + 2 = Monday.

Answer: Monday

Q3. What month comes 5 months after November?

Solution: Dec(1), Jan(2), Feb(3), Mar(4), Apr(5).

Answer: April

Q4. Arrange in logical order: Fruit, Seed, Tree, Sapling, Flower.

Solution: Seed → Sapling → Tree → Flower → Fruit.

Answer: Seed, Sapling, Tree, Flower, Fruit

12 Dozen & Eggs

12

KEY FORMULAS / FACTS

- 1 dozen = **12**; 1 gross = 144 (12 dozen); 1 score = 20
- Price per piece = Price per dozen / 12
- Price per dozen = Price per piece x 12
- Eggs left = Total - Broken/Sold
- Profit% = (SP - CP)/CP x 100 (per egg or per dozen - same %)
- **Hens-Eggs-Days:** Eggs = Hens x Days x (eggs per hen per day)

STANDARD METHOD

1. Convert everything into **pieces** (or everything into dozens).
2. Find cost per piece and selling price per piece.
3. Multiply by the number of pieces left after breakages.
4. Use the profit/loss formula on CP and SP.

SHORTCUT TRICK

The 12-Switch: Dozen to piece = ÷12; piece to dozen = ×12. For hen problems find the rate for **ONE hen in ONE day** first, then scale up.

Memory Hook: "D-12" - Dozen is Twelve. 'D' for Divide to get one piece.

Unit	Pieces / rate
1 dozen	12 pieces
1/2 dozen	6 pieces
2 dozen	24 pieces
1 gross	144 pieces (12 dozen)
1 score	20 pieces
Rs 60 per dozen	Rs 5 per piece

SOLVED EXAMPLES

Q1. 5 dozen eggs cost Rs 300. What is the cost of one egg?

Solution: 5 × 12 = 60 eggs. 300/60 = Rs 5.

Answer: Rs 5 per egg

Q2. Eggs bought at Rs 48 per dozen are sold at Rs 5 each. Find the profit percent.

Solution: CP each = 48/12 = 4. SP = 5. Profit = 1. Profit% = 1/4 × 100 = 25%.

Answer: 25%

Q3. A man buys 3 dozen eggs, 8 break, and he sells the rest at Rs 6 each. How much does he get?

Solution: 36 - 8 = 28. 28 × 6 = 168.

Answer: Rs 168

Q4. If 6 hens lay 6 eggs in 6 days, how many eggs will 12 hens lay in 12 days?

Solution: One hen lays 1 egg in 6 days. 12 hens x (12/6) = 24 eggs.

Answer: 24 eggs

13 Blood Relation

BR

KEY FORMULAS / FACTS

- Father's/Mother's brother = **Uncle**; Father's/Mother's sister = **Aunt**
- Father's father = Paternal grandfather; Mother's father = Maternal grandfather
- Sister's/Brother's son = **Nephew**; daughter = **Niece**
- Sister's husband / Wife's brother / Husband's brother = **Brother-in-law**
- Son's wife = Daughter-in-law; Daughter's husband = Son-in-law
- Uncle's/Aunt's children = **Cousins**
- Symbols: **+** male, **-** female, **=** married, **|** parent-child

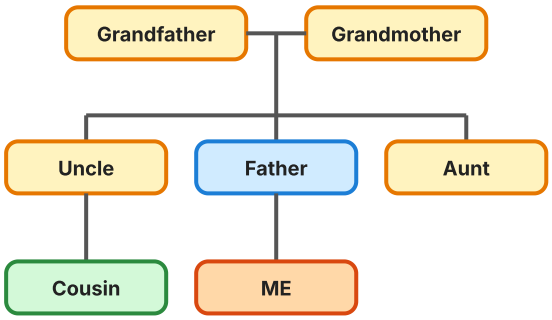
STANDARD METHOD

1. Start from the **end of the statement** and walk backward.
2. Replace 'my father's only son' with **me/brother**, etc.
3. Draw a small family tree; mark gender using symbols.
4. Read the final relation off the tree (look at generation and side).

SHORTCUT TRICK

Ladder Trick: Imagine 3 floors: **Grandparents / Parents & uncles-aunts / You & cousins**. Move up for parents, sideways for siblings, down for children. Read the answer from where you land.

Memory Hook: "Start from the End" - solve the last clue first, then work back.



Draw it first. Every relation becomes a line.

SOLVED EXAMPLES

Q1. Pointing to a man, Ali said, 'He is the son of my father's only son.' How is the man related to Ali?

Solution: My father's only son = Ali himself. So the man is Ali's son.

Answer: Son

Q2. A is B's sister; B is C's father. How is A related to C?

Solution: B is father of C, A is B's sister = Father's sister.

Answer: Aunt

Q3. Introducing a woman, a man said, 'Her mother is the only daughter of my mother.' How is the woman related to the man?

Solution: Man's mother's only daughter = Man's sister. Her mother is his sister so the woman is his sister's daughter.

Answer: Niece

Q4. Pointing to a photograph, Sara said, 'He is the brother of my father's son.' Who is he?

Solution: My father's son = my brother, so he is my brother's brother = my brother (or me if I am male).

Answer: Brother

14 Profit & Loss

P&L

KEY FORMULAS / FACTS

- Profit = SP - CP; Loss = CP - SP
- **Profit % = (Profit/CP) x 100; Loss % = (Loss/CP) x 100**
- SP = CP x (100 + P%)/100 (profit); SP = CP x (100 - L%)/100 (loss)
- CP = SP x 100/(100 + P%) (profit); CP = SP x 100/(100 - L%) (loss)
- Discount = Marked Price - SP; Discount% = (Discount/MP) x 100
- **Two items sold at same SP, one at x% profit and other at x% loss: always a LOSS of (x/10)%**
- Successive discounts a%, b%: Net = a + b - (ab/100)

STANDARD METHOD

1. Identify CP, SP, MP in the problem (CP = cost, SP = sold at, MP = marked).
2. Profit/Loss % is **always on CP**; discount % is **always on MP**.
3. Use the right formula for the unknown (CP, SP or %).
4. Check: is the answer a profit or loss as the problem says?

SHORTCUT TRICK

Fraction-Ratio Trick: 20% profit = 1/5 so CP:SP = 5:6; 25% = 1/4 → 4:5; 10% = 1/10 → 10:11; 50% = 1/2 → 2:3. For a loss use (n - 1) : n form.

Memory Hook: "CP is the King" - percent is always taken on Cost Price (and on Marked Price for discount).

%	Fraction	CP : SP (profit)	CP : SP (loss)
10%	1/10	10 : 11	10 : 9
12.5%	1/8	8 : 9	8 : 7
20%	1/5	5 : 6	5 : 4
25%	1/4	4 : 5	4 : 3
33.33%	1/3	3 : 4	3 : 2
50%	1/2	2 : 3	2 : 1

SOLVED EXAMPLES

Q1. CP = Rs 400 and SP = Rs 500. Find the profit %.

Solution: Profit = 100. 100/400 × 100 = 25%.

Answer: 25%

Q2. An article costing Rs 800 is sold at 15% loss. Find the SP.

Solution: SP = 800 × 85/100 = Rs 680.

Answer: Rs 680

Q3. After a 20% profit, an article is sold for Rs 1080. Find its CP.

Solution: CP = 1080 × 100/120 = Rs 900.

Answer: Rs 900

Q4. Two articles are sold at the same price, one at 10% profit and the other at 10% loss. Overall result?

Solution: Loss % = (10/10)% = 1%.

Answer: 1% loss

★ Last-Minute Cheat Sheet

Topic	One-line Formula / Trick
Age	Difference never changes; use ratio units; sum grows by n x people
Direction	N-E-S-W clockwise; Right = clockwise; distance = √(a²+b²); morning shadow = West
Percentage	Learn 1/2 ... 1/10 table; net = a + b + ab/100; 'x% more' - x/(100+x) less
Ranking	Total = Top + Bottom - 1; Between = |A - B| - 1
Coding	A=1 ... Z=26; opposite = 27 - position; EJOTY = 5,10,15,20,25
Mirror/Water	Mirror = 11:60 - T; Water = 6:00 - T; HIOX never change
Series	Check D-R-S-A: Difference, Ratio, Square/cube, Alternate
Analogy	Make a sentence from pair 1, test every option with it
Classification	Three share a property; find the stranger
Idioms	Never literal; picture + exaggerate; group by theme
Time Sequence	Odd days = mod 7; leap year 2, ordinary 1; Sun = 0 ... Sat = 6
Dozen & Eggs	1 dozen = 12; per piece = dozen price / 12; hens-days ratio
Blood Relation	Start from the end; draw a family tree; 3 floors ladder
Profit & Loss	On CP always; SP = CP(100+P)/100; equal SP at +x/-x% = loss (x/10)%

EXAM-DAY STRATEGY

- Attempt the **easy, scoring topics first**: Ranking, Coding, Direction, Series, Blood Relation.
- Do not spend more than **45-60 seconds** per question; skip and return.
- Always **draw** (family tree, direction sketch, number line).
- Revise the formula table above 10 minutes before the test.
- Practise daily: 20 questions mixed from all 14 topics.